

Problem 1: Understanding System Properties

Each of the following equations describes a relationship that holds between an input signal x and an output signal y of an associated discrete-time system, for all integers n . In each case, determine if the system is (i) memoryless, (ii) causal, (iii) BIBO stable.

Here are a few refreshers on definitions. Suppose a system maps $x[n]$ to $y[n]$.

- The system is said to be *memoryless* if the output at time n depends only on the input at time n . Intuitively, the present only depends on the present. Note that *all* memoryless systems are causal.
- The system is said to be *causal* if the output at time n depends only on inputs at or before time n . Intuitively, the present only depends on the present and the past - the system doesn't "look ahead".
- The system is said to be *Bounded-Input Bounded-Output (BIBO) stable* if every bounded input produces a bounded output. Formally, let x be any input to a system H such that $\max_n |x[n]| \leq B_x$ for some $B_x \in \mathbb{R}$. Then H is BIBO stable if and only if the output to x , call it y , satisfies $\max_n |y[n]| \leq B_y$ for some $B_y \in \mathbb{R}$. B_x, B_y are the "bounds" on x and y respectively, providing a (not necessarily tight) upper bound on their absolute values. For example, $x[n] = \delta[n]$ is bounded by 1, as well as by 2.

(a) $y[n] = 7x[n + 1]$

- (i) **Memoryless: No.** The output at time n does not depend solely on the input at time n (it depends on the input at time $n + 1$), so the system isn't memoryless. For example, $y(0)$ cannot be calculated from $x(0)$ - we need to know $x(1)$.
- (ii) **Causal: No.** The system "looks ahead", requiring the input at time $n + 1$ before it can determine the value at time n .
- (iii) **Stable: Yes.** Shifting a signal doesn't change its largest magnitude, and scaling by 7 just scales the bound by 7; the output is still bounded. For example, if $\max_n |x[n]| = B_x$, then $\max_n |y[n]| = 7B_x$.

(b) $y[n] = ax[n - n_0] + b$, where $a, b \in \mathbb{C}$, $n_0 \in \mathbb{Z}$. In cases where you answer no: does your answer become yes change if we fix $b = 0$? What if we fix $n_0 = 0$?

- (i) **Memoryless: No.** In general, $y[n]$ depends on $x[n - n_0]$. If $b = 0$, this doesn't affect anything. If $n_0 = 0$, the system is memoryless: the current output value is precisely the current input value, scaled by a , with b added to it.
- (ii) **Causal: Yes if and only if $n_0 \geq 0$.** If $n_0 < 0$, then we have to access a future value (e.g., $x[n + 1]$, $x(n + 2)$) to determine the output at time n .
- (iii) **Stable: Yes.** If the signal x is bounded by B_x , then y is bounded by $|aB_x + b|$.

The system here is an *affine* system: a linear system plus a constant offset. Unless the offset is zero, it's not a linear system, as a zero input must always map to a zero output.

(c) $y[n] = x[n]x[n - 1]$

- (i) **Memoryless: No.** For example, $y(0)$ depends on $x(0)$ as well as $x(-1)$.
- (ii) **Causal: Yes.** We only need the current input value, $x[n]$, and the one just prior to it, $x[n - 1]$, to determine the current output value $y[n]$.
- (iii) **Stable: Yes.** If x is bounded by B_x , then y is bounded by B_x^2 .

(d) $y[n] = e^{x[n]}$

- (i) **Memoryless: Yes.** The output at time n is precisely e to the power of the input at time n .
- (ii) **Causal: Yes.** All memoryless systems are causal.
- (iii) **Stable: Yes.** If x is bounded by B_x , then y is bounded by e^{B_x} .

(e) $y[n] = x(-n)$

- (i) **Memoryless: No.** For example, $y(5)$ depends on $x(-5)$.
 - (ii) **Causal: No.** For example, $y(-1)$ depends on $x(1)$, a future value.
 - (iii) **Stable: Yes.** If x is bounded by B_x , so is y .
- (f) $y[n] = v[n]x[n]$, where v is a fixed, *bounded* discrete-time signal.
- (i) **Memoryless: Yes.** To calculate the output at time n , just scale the input value at time n by the value $v[n]$.
 - (ii) **Causal: Yes.** All memoryless systems are causal.
 - (iii) **Stable: Yes.** If v is bounded by B_v , and x is bounded by B_x , then y is bounded by $B_v B_x$. Note that if v were unbounded, the system would not be stable.

The system here is ubiquitous in digital signal processing, where v is known as a *windowing function*. It's often of interest to crop signals and focus on one part of them by masking out all irrelevant samples as zero. By setting $v[n]$ to 1 at places we want to keep the signal (almost always, these will be consecutive samples), and 0 at places we want to discard, we can apply a rectangular window, for example. In EE 123, a follow on course to EE 120, the engineering tradeoffs involved in the design of v are discussed in more detail.

Problem 2: Dirac Delta and CT Convolution Practice

In discrete-time, we talked about the notion of an impulse response as a system's response to a Kronecker delta, $\delta(n)$. In continuous-time, we have a similar notion, but we must use the Dirac delta, $\delta(t)$. Similarly, continuous-time convolution is defined as

$$x * y(t) = \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \quad (1)$$

The Dirac delta has the following properties:

- $\delta(t)$ is zero $\forall t \neq 0$, and is infinite-valued at $t = 0$.
- $\int_A^B \delta(t) dt = 1$ if $A < 0 < B$ and 0 otherwise.
- **Sampling property:** $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$.
- **Sifting property:** $\int_A^B x(t)\delta(t - t_0) dt = x(t_0)$ if $A < t_0 < B$ and 0 otherwise.
- **Convolution property:** If $y(t) = \alpha\delta(t - t_0)$, then $(x * y)(t) = \alpha x(t - t_0)$.
- **Time-scaling property:** $\delta(at) = \frac{1}{|a|}\delta(t)$.

(a) Evaluate the following expressions.

(i) $\int_{-\infty}^{\infty} [3\delta(\tau) + 2\delta(\tau - 1)] d\tau$

$$3 \int_{-\infty}^{\infty} \delta(\tau) d\tau + 2 \int_{-\infty}^{\infty} \delta(\tau - 1) d\tau = 3 + 2 = 5 \quad (2)$$

(ii) $x(t) = \int_0^{\infty} \tau^2 \delta(t - \tau^2) d\tau$

Notice that our variable of integration, τ , is squared in argument of the Dirac delta. We don't have the tools to work with this directly (loosely speaking, this amounts to a non-constant time scaling), so we perform substitution on the integral.

If we define $u = \tau^2$, we get $du = 2\tau d\tau$, or $d\tau = \frac{du}{2\tau} = \frac{du}{2\sqrt{u}}$. In the range $\tau \in [0, \infty)$, τ^2 is an invertible function, so we can directly plug in this substitution to the bounds of integration.

$$x(t) = \int_0^\infty \frac{\sqrt{u}}{2} \delta(t - u) du = \int_0^\infty \frac{\sqrt{u}}{2} \delta(u - t) du.$$

In the second step, we applied a consequence of the time-scaling property, namely that the Dirac delta acts as an even function.

From here, we can apply the sifting property to get

$$x(t) = \begin{cases} \frac{\sqrt{t}}{2}, & t > 0 \\ 0, & t < 0 \end{cases}$$

(iii) $u * u(t)$, where $u(t)$ is the continuous-time unit step function:

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases} \quad (3)$$

$$u(t) * u(t) = \int_{-\infty}^{\infty} u(\tau) u(t - \tau) d\tau \quad (4)$$

Note that the product $u(\tau) u(t - \tau)$ is one iff both $u(\tau)$ and $u(t - \tau)$ are one. $u(\tau)$ is one if $0 \leq \tau$, and $u(t - \tau)$ is one if $t - \tau \geq 0$, or $\tau \leq t$. Thus, the integral is over a $1 \times t$ rectangle, starting from $t = 0$.

$$u(t) * u(t) = \begin{cases} t & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases} \quad (5)$$

Graphically, you are “sliding” two “boxes” towards one another as t increases. Once the “boxes” overlap, area accumulates at a linear rate indefinitely, yielding a “ramp.”

(b) Prove the convolution property of the Dirac delta, i.e., derive an expression for $x(t) * \alpha \delta(t - t_0)$, $\alpha, t_0 \in \mathbb{R}$, in terms of $x(t)$, where $x(t)$ is some continuous-time signal.

$$x(t) * \alpha \delta(t - t_0) = \int_{-\infty}^{\infty} x(\tau) \alpha \delta(t - t_0 - \tau) d\tau \quad (6)$$

$$= \alpha \int_{-\infty}^{\infty} x(t - t_0) \delta(t - t_0 - \tau) d\tau \quad (7)$$

$$= \alpha x(t - t_0) \int_{-\infty}^{\infty} \delta(t - t_0 - \tau) d\tau \quad (8)$$

$$= \alpha x(t - t_0) \quad (9)$$

Convolving an arbitrary continuous-time signal $x(t)$ with a scaled and shifted Dirac delta scales and shifts $x(t)$, just like in discrete-time. The Dirac delta, like the Kronecker delta, is the convolution identity.